

PAPER_043: The UQFF 26-Level Polynomial Energy Hierarchy: From Sub-Quantum Fluctuations to Universal Scales

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Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

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Validator: QCalc_Phase1_Validation.py (Test 1: PASS ?), test_phase2_validation.py (26/27 PASS)

Source Modules: QuantumLevel26Framework.py (630 lines), source172.cpp (SOURCE115)

Index Slot: 1.6 26-Dimensional Energy Structure,

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Abstract

The UQFF 26-level energy hierarchy provides a unified mathematical description of physical phenomena from the deepest quark confinement scale (10^{-18} m , $\sim 10^{-12} \text{ J}$) to the observable universe (10^6 m , $\sim 10^6 \text{ J}$). This paper establishes the precise mathematical foundation of this hierarchy through two complementary representations: the **polynomial energy formula** $E_n = 10^{(n-20)} \text{ J}$ (validated by QCalc_Phase1_Validation.py, Test 1 PASS) and the **vacuum density formula** $\rho_n = \rho_{\text{SCm}} n \bar{J}/\text{m}$ (validated by QuantumLevel26Framework.py). The Universal Inertia coupling operator U_{i_level} connects all 26 levels through the LENR resonance frequency $\tau_{\text{LENR}} = 1.25 \times 10^{-16} \text{ Hz}$. The core UQFF gravity equation $g(r,t) = S \tau_{\text{LENR}}^2 [Ug1_i + Ug2_i + Ug3_i + Ug4_i]$ emerges naturally from this hierarchical foundation.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The 26-Level Energy Hierarchy: Two Representations

1.1 Polynomial (Absolute Energy) Representation

The QCalc Phase 1 validator establishes the following absolute energy per level:

$$E_n = 10^{n-20} \text{ J}, \quad n = 1, 2, \dots, 26$$

Validation checkpoints (QCalc_Phase1_Validation.py Test 1 PASS ?): - $E_1 = 10^{-12} \text{ J}$ (sub-quantum fluctuations at quark scale) - $E_8 = 10^{-1} \text{ J}$ (nuclear binding, proton-neutron pairs) -

E18 = 10² J (Higgs boson energy scale) - E20 = 10 = 1 J (galactic vacuum, Ug4 reference)
 - E26 = 10⁶ J = 1 MJ (universal cosmological scales)

This representation spans **25 orders of magnitude** (10⁵ J total range), confirmed by the validator: Total Span = 1.0000e+25.

Each level is separated by exactly **one order of magnitude (10)**. This geometric spacing means level n covers a distinct energy decade, providing non-overlapping coverage of all known physical processes.

1.2 Vacuum Density (Local Field) Representation

The QuantumLevel26Framework module defines level energy densities via quadratic scaling:

$$\rho_n = \rho_{\text{SCm}} \times n^2, \quad \rho_{\text{SCm}} = 10^{-8} \text{ J/m}^3$$

This gives a parabolic energy density profile across the 26 levels. Unlike the polynomial representation (which is global and absolute), the density representation is local describing the vacuum energy density associated with quantum processes at each scale.

1.3 Complete 26-Level Table

Level	State Description	ρ_n (J/m ³)	E_n (J)	Scale (m)	ρ_i	Physical Examples
1	Quarks	1.00×10 ⁻⁸	1×10 ^{??}	10 ^{??8}	1.00	Quark confinement, pion exchange
2	Sub-nuclear shell	4.00×10 ⁻⁸	1×10 ^{??8}	10 ^{??7}	0.98	Nuclear binding, residual strong force
3	Nuclear quantum shell	9.00×10 ⁻⁸	1×10 ^{??7}	10 ^{??6}	0.95	Magic numbers, shell model
4	Nucleon pairing	1.60×10 ⁻⁷	1×10 ^{??6}	10 ^{??5}	0.93	Deuteron binding, spin coupling
5	Inner e ⁻ shells (K,L)	2.50×10 ⁻⁷	1×10 ^{??5}	10 ^{??4}	0.90	1s, 2s orbitals, X-ray transitions
6	Middle e ⁻ shells (M,N)	3.60×10 ⁻⁷	1×10 ^{??4}	10 ^{??}	0.88	3s, 3p, 3d orbitals, UV transitions
7	Outer e ⁻ shells (O,P,Q)	4.90×10 ⁻⁷	1×10 ^{??}	10 ^{??}	0.85	Valence electrons, visible light
8	Van der Waals	6.40×10 ⁻⁷	1×10 ^{??}	10 ^{??}	0.82	London dispersion, molecular binding
9	Molecular orbital	8.10×10 ⁻⁷	1×10 ^{??}	10 ^{??}	0.80	Covalent bonds, HOMO-LUMO gap
10	SOLIDS	1.00×10⁻⁶	10^{??}	10^{???}	0.75	Crystalline solids, proton mass, phonons
11	LIQUIDS	1.21×10⁻⁶	10^{???}	10^{??8}	0.70	Water, electron density waves

Level	State Description	ρ_n (J/m)	E_n (J)	Scale (m)	λ_i	Physical Examples
12	GASES	1.44×10^{-6}	10^{28}	10^{27}	0.65	Air molecules, ideal gas
13	PLASMA	1.69×10^{-6}	10^{27}	10^{26}	0.60	Solar corona, Langmuir waves
14	Molecular clusters	1.96×10^{-6}	10^6	10^{-5}	0.55	Proteins, colloids
15	Cellular structures	2.25×10^{-6}	10^{-5}	10^{-4}	0.50	Membranes, organelles
16	Macroscopic matter	2.56×10^{-6}	10^{-4}	$10^?$	0.45	Dust grains
17	Centimeter objects	2.89×10^{-6}	$10^?$	$10^?$	0.40	Rocks, organisms
18	Meter-scale	3.24×10^{-6}	$10^?$	10	0.35	Buildings, trees
19	Geological (km)	3.61×10^{-6}	$10^?$	10	0.30	Mountains, lakes
20	Planetary	4.00×10^{-6}	1 J	10^6	0.25	Earth, Moon, Mars (Ug4 anchor)
21	Stellar	4.41×10^{-6}	10^8 J	$10^?$	0.20	Sun, red dwarfs, white dwarfs
22	Solar system	4.84×10^{-6}	10	10	0.15	Heliosphere, Kuiper belt
23	Interstellar	5.29×10^{-6}	10	10^5	0.12	Nebulae, star clusters
24	Galactic	5.76×10^{-6}	10^4	10^8	0.10	Spiral arms, galactic disk
25	Supercluster	6.25×10^{-6}	10^5	10	0.08	Galaxy groups, Laniakea
26	Universal	6.76×10^{-6}	10^6 J	10^6	0.05	Observable universe, Hubble volume

2. Universal Inertia Coupling

2.1 Definition

The Universal Inertia at level i:

$$U_{i,\text{level}} = \lambda_i \cdot \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} \cdot \omega_{\text{LENR}} \cdot \cos(\pi t_n) \cdot (1 + f_{\text{TRZ}})$$

where: - λ_i = level-dependent coupling constant (Table above, column λ_i) - $\rho_{\text{SCm}}/\rho_{\text{UA}} = 10^{28}/10^? = \mathbf{10}$ (vacuum density ratio) - $\omega_{\text{LENR}} = 1.25 \times 10$ Hz (LENR resonance frequency) - t_n = negative time parameter (cosine modulation) - f_{TRZ} = time-reversal zone factor (default 0.01)

2.2 Level-10 Reference Value

For the solid-state reference (level 10, $t_n = 0$, $f_{\text{TRZ}} = 0.01$):

$$U_{i=10} = 0.75 \times 10^3 \times 1.25 \times 10^{12} \times 1.0 \times 1.01 = 9.47 \times 10^{14} \text{ J/m}^3 \cdot \text{Hz}$$

Validator confirms: Universal Inertia Level 10 ? PASS ?

3. Core UQFF Gravity Equation

The gravitational field at position (r, t) is the 26-layer superposition:

$$\mathbf{g}(r, t) = \sum_{i=1}^{26} [U_{g1,i}(r) + U_{g2,i}(r) + U_{g3,i}(r, t) + U_{g4,i}(r, t)]$$

where each contributes a distinct physical mechanism: - **Ug1_i**: Magnetic dipole buoyancy (SOURCE52): $U_{g1,i} = (E_{\text{DPM}}/r) \cdot f_{\text{TRZ}}$ - **Ug2_i**: Charge-reactivity (SOURCE54): $U_{g2,i} = s_{\text{field}} [UA]_i r$ - **Ug3_i**: String rotation (SOURCE56): $U_{g3,i} = O_{\text{string}} \cdot \sin(ip/26)$ - **Ug4_i**: Vacuum concentration (SOURCE57): $U_{g4,i} = M_{\text{source}} \cdot \text{vac}/(d E_{\text{LEP}})$

4. Dual Consistency of the Two Representations

The polynomial ($E_n = 10^{(n-20)}$) and density ($\rho_n = \rho_{\text{SCm}} n$) representations are related through the characteristic volume V_n at each level:

$$E_n = \rho_n \times V_n = \rho_{\text{SCm}} \times n^2 \times V_n = 10^{n-20} \text{ J}$$
$$\Rightarrow V_n = \frac{10^{n-20}}{\rho_{\text{SCm}} \times n^2} = \frac{10^{n-20}}{10^{-8} \times n^2} = \frac{10^{n-12}}{n^2} \text{ m}^3$$

This defines the **characteristic volume** at level n the volume over which the polynomial energy is distributed at the local vacuum density. For level 10: $V_{10} = 10^?/(100) = 10^{-4} \text{ m}^3$ a cube of side $\sim 0.046 \text{ m}$ (4.6 cm), consistent with the $10^? \text{ m}$ typical scale 10 lattice sites in a mole of solid.

5. Nuclear Binding Energy Check

Level 8 provides an observable verification point. The validator reports: - $E_8 = 10^? \text{ J} = 6.25 \text{ MeV}$ - Expected nuclear binding per nucleon: 8 MeV - Error: 21.97% (within 50% tolerance)

Validator: Test 1 Nuclear Binding Check ? PASS ? (at 21.97% error < 50% tolerance)

This 22% discrepancy at level 8 reflects the difference between the UQFF polynomial (purely geometric/exponential) and the QCD-derived nuclear binding energy. The UQFF 26-level polynomial is an energy scale index, not a precision nuclear physics formula but it correctly locates level 8 within the nuclear binding energy decade ($10^? \text{ J}$ 6 MeV).

Conclusions

The UQFF 26-level energy hierarchy provides a self-consistent, geometrically structured energy index spanning 25 decades. Both the polynomial ($E_n = 10^{n-20}$) and density ($\rho_n = \rho_{SCM} n$) representations are validated. Levels 1013 correspond to the four classical matter states (solid/liquid/gas/plasma), with level 20 anchoring the Ug4 galactic vacuum scale and level 26 marking the observable universe boundary.

Validators: `QCalc_Phase1_Validation.py` Test 1 PASS ? | `test_phase2_validation.py` 26/27 PASS | $\rho = 0.0005/\text{day}$ | $[SSq] = 0.57$

Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60-0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	<code>compute_Ug1_SOURCE4 / compute_Ug1()</code>
Ug2	Outer-field bubble (charge-reactivity)	<code>compute_Ug2_SOURCE4 / compute_Ug2()</code>
Ug3	Magnetic string rotation	<code>compute_Ug3_SOURCE4 / compute_Ug3()</code>
Ug4	Vacuum concentration (star-BH)	<code>compute_Ug4_SOURCE4 / compute_Ug4()</code>
Ubi	Buoyancy force	<code>compute_Ubi_SOURCE4 / compute_Ubi()</code>
Um	Universal Magnetism (Heaviside-amplified)	<code>compute_Um_SOURCE4 / compute_Um()</code>
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	<code>compute_FU_SOURCE4 / full pipeline</code>

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times (1 + 10^{13} \Theta(\rho_{\text{SCm}} - \rho_c)) \times (1 + A_q \cos(\Delta\omega t))$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times \text{Ubi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.